

ISHAN UDAWATTE

Software Engineering Undergraduate | Backend & Full-Stack Developer

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PROFESSIONAL SUMMARY

Software Engineering undergraduate at the Informatics Institute of Technology (University of Westminster) with hands-on experience building and deploying production-grade systems. Delivered AquaSense, a live IoT water monitoring platform handling real-time device communication, REST APIs, and a Flutter mobile app. Skilled in backend architecture, database design, and full-stack development across Python, Java, JavaScript, and PHP ecosystems. Actively seeking an internship where I can contribute to scalable software while continuing to grow in a professional engineering environment.

TECHNICAL SKILLS

Area	Technologies
Languages	Python, Java, JavaScript, PHP, C++, Dart
Backend	FastAPI, Node.js, Express.js, REST API Design, JWT Authentication, SQLAlchemy ORM
Databases	PostgreSQL, MySQL, MongoDB, Firebase, Supabase
Frontend	React, EJS, HTML5, CSS3, Flutter (Dart)
IoT & Embedded	ESP32, MQTT (HiveMQ), IoT Data Processing
Tools & Platforms	Git, GitHub, NetBeans, Postman, XAMPP
Methodologies	Agile / Scrum, OOP, UML, Test-Driven Development (JUnit)
ML & Data	Machine Learning Fundamentals, Data Processing

PROJECTS

AquaSense - Smart IoT Water Monitoring & Leak Detection System | Live Production Deployment 2025 – 2026

- Built and deployed a full-stack IoT platform that monitors real-time water usage and detects leaks across multiple zones, currently serving active users.
- Developed ESP32 firmware in C++ for real-time water flow monitoring, automated valve control, leak detection with false-positive prevention, heartbeat monitoring, and MQTT-based secure cloud integration with real-time alert generation.
- Engineered 12+ FastAPI (Python) REST endpoints for device communication, analytics queries, and mobile app integration; reduced average API response time through async processing.
- Designed a PostgreSQL schema with SQLAlchemy ORM covering device telemetry, user accounts, consumption analytics, and automated leak-alert logging.
- Implemented secure JWT-based authentication and role-based user management; integrated MQTT (HiveMQ broker) for sub-second ESP32 - server telemetry.
- Delivered a Flutter mobile application for live dashboards, historical consumption charts, and instant leak notifications.
- Practised full Agile workflow: sprint planning, Git branching strategy, peer code reviews, and continuous deployment

Technologies: FastAPI, Python, C++, PostgreSQL, SQLAlchemy, Flutter, MQTT (HiveMQ), ESP32, Firebase, Git

Acyclicity Checker - Directed Graph Algorithm Implementation | Individual Project 2026

- Implemented a sink-elimination algorithm ($O(V+E)$) on a custom adjacency-list graph structure using dual HashMaps for out-edges and in-edges, enabling $O(1)$ sink detection and predecessor lookups without full-graph rescans.
- Built an iterative DFS cycle detector with three-colour marking (White/Gray/Black) to find and reconstruct concrete cycle paths; iterative design avoids Java stack overflow on large inputs.
- Identified and fixed a predecessor-map corruption bug in cycle path reconstruction, documenting the root cause and fix with clear commentary.
- Empirically validated linear time complexity across graph sizes up to 320 nodes, with doubling ratios consistently below 2.0, confirming $O(V+E)$ behaviour.

Technologies: Java, OOP, Graph Algorithms, Data Structures, JUnit

Smart Home E-Commerce Web Application | Individual Project 2026

- Built a multi-page PHP/MySQL e-commerce platform for a smart home retailer, covering product browsing, a session-based shopping basket, user authentication, and role-based navigation.
- Implemented a session-managed basket with add, remove, quantity selection, and clear functionality, with live subtotal and total calculations from database queries.
- Designed a relational MySQL database schema for products (inventory, pricing, images) and users, accessed via mysqli.
- Built dynamic navigation that adapts based on login state, showing guest vs. member vs. staff options using PHP session variables.
- Applied consistent UI across all pages using a custom CSS stylesheet with responsive layout, Google Fonts, and hover transitions.

Technologies: PHP, MySQL, HTML5, CSS3, JavaScript, Apache (XAMPP)

To-Do List Web App | Individual Project 2026

- Built a full-stack CRUD web application using Node.js and Express, with MongoDB (via Mongoose) for persistent task storage.
- Implemented RESTful routes for creating, reading, updating, and deleting tasks, with EJS server-side templating for dynamic rendering.
- Used Mongoose schema modelling with default timestamps and environment-based database configuration via dotenv.

Technologies: Node.js, Express.js, MongoDB, Mongoose, EJS, HTML5, CSS3

University Enrollment Management System | Individual Project 2025 – 2026

- Extended a multi-class Java system (Student, Lecturer, Person, EnrollmentManager) by implementing new features and refactoring for maintainability.
- Applied OOP principles - inheritance, polymorphism, and encapsulation - and validated correctness with a JUnit test suite.
- Redesigned Java Swing GUI components to meet updated functional requirements and improve usability.

Technologies: Java, OOP, NetBeans, Java Swing, Collections Framework, JUnit, UML

EDUCATION

BEng (Hons) Software Engineering | University of Westminster (via IIT Sri Lanka) 2024 – 2028

- Level 5 : Object Oriented Programming, Database Systems, Algorithms Theory Design & Implementation, Server Side Web Development, Software Engineering Principles & Practice, Software Development Group Project
- Level 4 : Mathematics for Computing, Software Development, Computer Systems Fundamentals, Trends in Computer Science, Web Design and Development

G.C.E. Ordinary Level | D.S. Senanayake College 2019 – 2022

- 4 A passes, 3 B passes, 2 C passes (Mathematics, English, I.C.T, Science, Business & Accounting, Arts, Sinhala, History, Buddhism)

CERTIFICATIONS

Informatics Institute of Technology

- Professional Certificate in Machine Learning 32 Hours Programme (Merit)
- Foundation Certificate in Higher Education Completed

LEADERSHIP & EXTRACURRICULAR

- Head Prefect, D.S. Senanayake College (2022/23)
- Vice-Captain, D.S. Senanayake College 1st XV Rugby Team
- Chief Organizer, Aeronautical Society
- Sergeant at Arms, Leo Club

REFERENCES

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I hereby certify that all the above-furnished particulars are true and correct to the best of my knowledge and belief.



Ishan Udawatte